



AI from Zero to Hero

Ahadu Bank

Data Science Project

Overview

This project focuses on performing Exploratory Data Analysis (EDA) on the historical credit dataset provided for the **Credit Scoring**. The aim of this analysis is to support inclusive finance and data-driven lending decisions in Ethiopia by identifying key risk factors that predict the likelihood of a borrower experiencing financial distress within the next two years. The insights will directly inform the development of a high-performing credit scoring model and its packaging as a scalable Model-as-a-Service (MaaS) API.

Dataset:

The dataset used in this analysis contains anonymized historical data from **150,000 borrowers** (training set provided for the challenge). Below is a complete summary of **all columns** in the dataset:

- **Unnamed: 0:** Unique borrower ID (row index)
- **SeriousDlqin2yrs:** Target variable – 1 if the borrower experienced financial distress in the next 2 years, 0 otherwise
- **RevolvingUtilizationOfUnsecuredLines:** Total balance on unsecured credit lines divided by total credit limits (ratio, values >1 indicate over-utilization)
- **age:** Age of the borrower



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- **NumberOfTime30-59DaysPastDueNotWorse:** Number of times the borrower was 30–59 days past due (but not worse) in the last 2 years
- **DebtRatio:** Monthly debt payments divided by monthly gross income
- **MonthlyIncome:** Monthly gross income of the borrower
- **NumberOfOpenCreditLinesAndLoans:** Number of open credit lines and loans
- **NumberOfTimes90DaysLate:** Number of times the borrower was 90 days or more past due
- **NumberRealEstateLoansOrLines:** Number of mortgage and real-estate loans (including home-equity lines)
- **NumberOfTime60-89DaysPastDueNotWorse:** Number of times the borrower was 60–89 days past due (but not worse) in the last 2 years
- **NumberOfDependents:** Number of dependents (excluding self)

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